

Automated Disease Diagnosis from Symptom Descriptions Using NLP Techniques

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Abstract—Building machine learning models to classify diseases from symptom descriptions holds promise for augmenting diagnostic workflows, particularly in low-resource or remote settings. However, many existing models rely on clean, structured, and short-text symptom inputs, which poorly reflect how real patients express symptoms in clinical contexts. This study presents a hybrid natural language processing (NLP) pipeline designed to classify 24 diseases based on free-text, human-like symptom descriptions.

We began by cleaning and standardizing a curated dataset, then applied traditional feature extraction (TF-IDF) and classical machine learning models (Logistic Regression, Random Forest). While these approaches yielded strong validation accuracy (up to 98%), they failed to generalize to Simulated patient-style evaluation symptom phrasing. To bridge this gap, we transitioned to contextual embeddings using FastText and transformer-based models (BERT, ClinicalBERT, PubMedBERT, BioBERT, and RoBERTa), and enhanced performance further by integrating Named Entity Recognition (NER) outputs into the classification pipeline.

A final dataset augmentation step introduced 240 new patient-style symptom descriptions, generated by large language models (LLMs) and verified by a clinical expert. Our best-performing model, RoBERTa with Text + NER input, achieved 96% validation accuracy, 95% test accuracy, and 100% Top-3 accuracy on Simulated patient-style evaluation cases. The complete pipeline was deployed in a web interface, demonstrating potential for clinical applications. These findings highlight the importance of semantic understanding, structured extraction, and language variation in improving AI-based diagnostic tools. Clinical validation with real patient data is required before deployment.

I. INTRODUCTION

Accurate and timely disease diagnosis remains a persistent challenge in modern healthcare, especially when reliant on symptom descriptions provided directly by patients. Patient-reported symptoms vary widely in structure and clarity, influenced by health literacy, culture, and communication style. Such variability can lead to misdiagnoses or delays in treatment, particularly when systems are not equipped to interpret nuanced or non-standard expressions of symptoms [1].

Natural Language Processing (NLP), a subfield of artificial intelligence, has emerged as a powerful tool for extracting meaningful patterns from textual data. Its applications in healthcare are growing, including tasks such as medical record mining [2], automated symptom classification [3], and patient-facing chatbots for triage and consultation [4]. These innovations point toward a future where AI not only supports but

could potentially support diagnostic workflows, especially in primary care, emergency triage, or low-resource environments.

This research focuses on building a robust and generalizable model for predicting diseases from free-text symptom descriptions, using a blend of traditional and deep learning-based NLP techniques. Unlike prior approaches that rely on short, structured symptom inputs, we aim to model real-world, patient-style narratives, those reflecting informal, unstructured language that challenges standard classification systems.

Our initial approach involved standard text Preprocessing, Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction [5], and classical models like Logistic Regression and Random Forest. While these models achieved high accuracy on validation and test sets (up to 98%), they failed to generalize when tested on realistic symptom narratives generated using large language models (LLMs) [6]. To bridge this gap, we transitioned to FastText embeddings [7] and ultimately adopted transformer-based architectures such as BERT [8], BioBERT [9], ClinicalBERT [10], PubMedBERT [11], and RoBERTa [12]. We further integrated (NER) using biomedical tools such as SciSpacy [13] as a structured signal enhancer and augmented our dataset with medically accurate, human-like symptom variations to improve robustness.

Our final model, RoBERTa with both text and NER inputs, achieved 96% validation accuracy, 95% test accuracy, and 100% Top-3 accuracy on Simulated patient-style evaluation cases. This model was deployed in a fully operational web interface, demonstrating real-world applicability and readiness for potential healthcare applications.

This paper details our full pipeline, from data cleaning and augmentation to model experimentation and deployment, emphasizing the practical challenges of generalization, semantic understanding, and clinical usability. In doing so, we aim to contribute a scalable and patient-centered approach to AI-assisted disease diagnosis.

II. RELATED WORK

The application of Natural Language Processing (NLP) to healthcare has seen rapid growth over the past decade, driven by increasing availability of clinical text and advancements in language modeling. One of the most relevant areas is symptom-based disease prediction, where the goal is to in-

interpret patient-reported symptoms and classify the underlying condition.

Classical NLP Approaches: Early work in symptom classification often relied on rule-based systems or keyword matching, which were limited in flexibility and generalizability [3]. Later methods leveraged traditional NLP pipelines involving tokenization, stopword removal, and statistical representations such as Bag-of-Words or Term Frequency-Inverse Document Frequency (TF-IDF) [5]. These representations, while interpretable and computationally efficient, treat text as unordered token counts and fail to capture deeper semantic relationships between words. Models such as Logistic Regression and Support Vector Machines (SVM) were commonly used on such features and yielded competitive performance on structured or cleaned datasets [14].

Word Embeddings and Lightweight Neural Models: To capture richer semantic information, distributed word representations like Word2Vec and FastText were introduced [7]. FastText, in particular, was shown to perform well on short clinical texts due to its subword modeling capabilities. When coupled with classifiers such as Random Forests or shallow neural networks, embedding-based models outperformed TF-IDF in tasks where symptom variation and word overlap were minimal. However, their performance still suffered in generalizing to unseen linguistic patterns or idiomatic patient language [15].

Transformer Models in Medical Text Classification: The emergence of transformer-based models like BERT [8] and its medical variants, BioBERT [9], ClinicalBERT [10], PubMedBERT [11], RoBERTa [12], and BioMedBERT [16] marked a turning point in medical NLP. These models leverage attention mechanisms and deep contextual representations to understand the full semantics of clinical narratives. Studies have shown significant improvements in tasks such as Named entity recognition, ICD coding, and medical question answering. However, their application to patient-style symptom narratives remains relatively underexplored, especially in the context of disease classification from free-text inputs.

Named Entity Recognition in Clinical NLP: NER plays a crucial role in extracting structured information from unstructured text. Systems like cTAKES [17] and SciSpaCy [13] were developed specifically for clinical texts but often struggle with casual language, patient misspellings, and non-standard phrasing. Transformer-based NER pipelines, such as those built using Hugging Face models (e.g., *d4data/biomedical-ner-all*), have shown higher adaptability to noisy or layperson-generated symptom descriptions. When used as auxiliary features in classification tasks, structured symptom entities can improve both model interpretability and robustness.

Data Augmentation in Medical NLP: Due to the sensitive nature of clinical data and challenges in annotation, many recent works have turned to data augmentation using Large Language Models (LLMs) such as ChatGPT or Claude [6]. While synthetic data must be used with caution, studies suggest that carefully generated and clinically verified augmentations can significantly improve generalization, especially when models

are deployed in real-world settings where language diversity is high.

Positioning of This Work: This study builds on the above foundations by integrating traditional, embedding-based, and transformer-based techniques in a single pipeline, and evaluating their ability to handle human-like symptom expressions. It further distinguishes itself through the use of structured NER features and Simulated patient-style evaluation, an area where current literature is sparse. Our findings highlight the limitations of classical approaches and demonstrate how a carefully augmented and semantically enriched model, such as RoBERTa with NER, can bridge the gap between lab performance and real-world usability.

III. DATASET AND PREPROCESSING

A. Dataset Description

The dataset used in this study is derived from the publicly available Symptom2Disease corpus [18], consisting of 1,200 samples equally distributed across 24 disease categories (50 samples per disease) As shown in Figure 1:

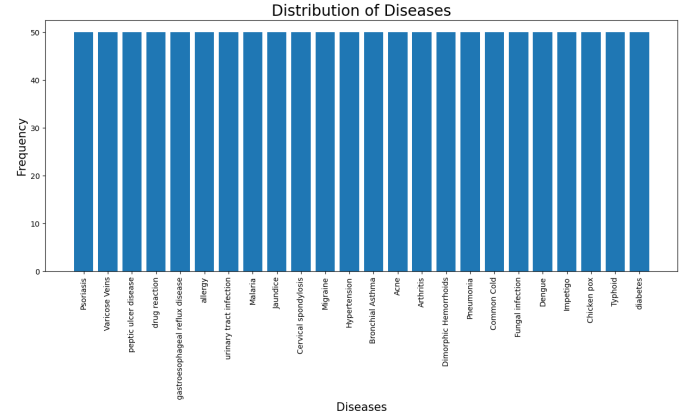


Fig. 1. Sample distribution across the 24 diseases in the Symptom2Disease dataset, with uniform counts per category to maintain a balanced dataset for training and evaluation.

Each entry contains a free-text symptom description with a corresponding disease label, ensuring a balanced distribution across classes.

To enhance model robustness, 240 additional samples, ten per disease, were generated using AI models (ChatGPT, Claude, Gemini, DeepSeek). The AI was instructed to produce realistic patient-style symptom descriptions based on medical literature (e.g., UpToDate, Mayo Clinic), covering primary and secondary symptoms, severity, temporal progression, and natural patient language. All entries were manually reviewed by a medical expert, resulting in a final dataset of 1,440 samples

B. Data Preprocessing

The preprocessing phase aimed to transform raw symptom descriptions into a consistent and machine-readable format, while preserving their clinical semantics. Several carefully designed steps were applied, as detailed below:

a) Lowercasing: All symptom texts were converted to lower-case to avoid mismatches due to capitalization. For instance, “Fever” and “fever” were treated identically.

b) Noise and Punctuation Removal: All special characters were removed from the text using a custom regular expression. Punctuation was selectively handled, commas and periods were retained initially to assist in entity parsing and later stripped before vectorization. Digits were removed after verifying that no useful numeric indicators existed in the dataset.

c) Contraction Handling: During early experiments, tokenization incorrectly split contractions (e.g., “I’ve” → “I,m”). A post-processing function was applied to map common fragments such as “ve” → “have” and “m” → “am”, restoring grammatical integrity for better semantic processing.

d) Misspelling Correction: Spelling inconsistencies were corrected using the TextBlob library [19]. For example, “abdoman” was corrected to “abdomen.” This step improved token consistency and reduced embedding sparsity.

e) Tokenization: Each symptom description was broken into word-level tokens using SpaCy’s `en_core_web_sm` model. Tokenization enables further steps such as stopword filtering and lemmatization to operate on individual components of the text.

f) Stopword Handling: Common English stopwords were considered for removal. However, unlike general-purpose NLP pipelines, we retained clinically relevant functional words such as “have”, “also”, “has”, and “in” to preserve diagnostic context. A custom stopword filter was implemented to strike a balance between noise reduction and clinical expressiveness.

g) Lemmatization: Using SpaCy, each word was reduced to its base or dictionary form. For instance, “vomiting” and “vomited” were both lemmatized to “vomit.” This reduces lexical variation while maintaining semantic clarity.

h) Post-Lemmatization Clean-Up: A final pass reassembled the cleaned tokens into full sentences, ensuring a consistent format for downstream vectorization and modeling.

Figure 2 illustrates the dominance of high-frequency stopwords in the raw dataset. This observation motivated the combined application of stopword filtering and lemmatization to focus the model on informative terms.

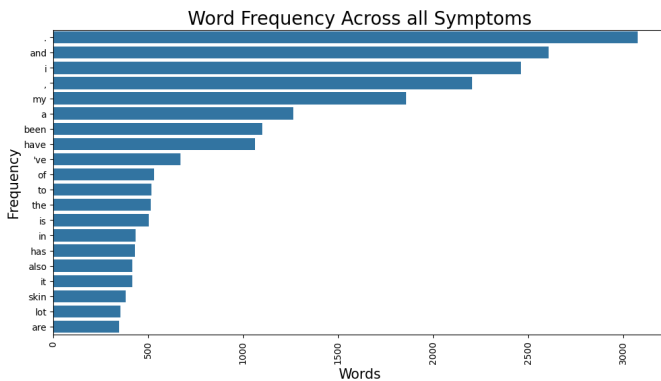


Fig. 2. Word frequency distribution across all symptom descriptions. Common stopwords dominate, justifying the need for preprocessing steps like stopword filtering and lemmatization.

C. Final Cleaned Text

Following all transformations, the symptom descriptions were cleaned, normalized, and reassembled. Inspection revealed that most entries ranged from 12 to 55 words, with a mean length of 30.7 words. The final dataset was saved for subsequent feature engineering and (NER) extraction.

IV. FEATURE ENGINEERING

This section describes the techniques used to transform raw symptom descriptions into numerical representations suitable for classification. Three primary approaches were explored: TF-IDF, word embeddings using FastText, and transformer-enhanced inputs augmented with biomedical (NER).

A. TF-IDF Vectorization

We initially employed Term Frequency–Inverse Document Frequency (TF-IDF) as a classical vectorization method [14]. This approach represents each symptom description as a weighted vector, emphasizing words that are frequent within a document but rare across the corpus. To reduce dimensionality and improve generalization, Truncated Singular Value Decomposition (SVD) was applied, compressing the TF-IDF vectors to 300 dimensions. These transformed vectors were subsequently used with classical machine learning models, including Logistic Regression and Random Forest. While this approach achieved high validation accuracy (96%), it failed to generalize to Simulated patient-style text variations, correctly predicting only 2 out of 6 evaluation samples. The poor performance was attributed to the sparsity of the representation (99.15% zeros) and its sensitivity to exact word matches, which limited robustness to variations in symptom phrasing.

B. Word Embeddings with FastText

To overcome the semantic limitations of TF-IDF, we experimented with FastText embeddings [7], which represent words as averages of character-level n-grams. This allows the model to capture morphological variations and out-of-vocabulary terms.

Two configurations were tested:

- **Custom FastText:** Trained on the symptom dataset using 100-dimensional vectors, 5 epochs, and a window size of 5.
- **Pretrained FastText:** Used 300-dimensional pretrained vectors from the `cc.en.300.vec` model, which is trained on the Common Crawl corpus using subword information. These vectors were loaded using the Gensim library’s `load_word2vec_format()` function and later converted to a binary key-value format for faster access during training.

These embeddings were averaged per sentence to generate fixed-length vectors. While they provided some improvement in semantic sensitivity, real-world generalization remained limited, with both models achieving only 2 out of 6 correct predictions on external symptom samples.

C. Named Entity Recognition (NER)

To address the generalization gap further, we introduced a biomedical (NER) pipeline using the transformer-based model `d4data/biomedical-ner-all`.

Entity Extraction: NER was applied to each symptom description to extract relevant medical concepts such as diseases, symptoms, body parts, and procedures.

Cleaning and Normalization: Extracted entities were cleaned to remove generic or fragmented terms (e.g., “increase,” “problem”) and corrected for spelling inconsistencies. Context-specific mappings were applied (e.g., “spicy” → “spicy food intolerance”).

NER as Input: The final NER output was either:

- Used alone as input (NER only), or
- Concatenated with the original cleaned text (Text + NER), forming a dual-input representation.

This dual-input format provided the model with both natural phrasing and structured biomedical context. The best results were achieved using the **Text + NER** configuration, especially with transformer-based models (see Table I for examples of extracted entities).

V. MODELING APPROACHES

This section outlines the classification models tested during the project, ranging from traditional machine learning to transformer-based deep learning architectures. Each model was evaluated using three types of input: (1) symptom text only, (2) (NER) output only, and (3) a combined format of symptom text concatenated with NER entities (Text + NER).

A. Classical Models

We began with two classical machine learning classifiers using TF-IDF + SVD-reduced vectors:

- **Logistic Regression:** A regularized linear model known for its simplicity and interpretability. We applied L2 regularization and tuned the penalty strength hyperparameter (C).
- **Random Forest:** An ensemble of decision trees designed to reduce overfitting. Hyperparameters such as the number of estimators and maximum tree depth were tuned using validation accuracy.

While both models achieved high validation accuracy (95–96%), they demonstrated poor generalization to real-world symptom phrasing, with Top-1 accuracy as low as 33% on external examples.

B. Embedding-Based Models (FastText)

Next, we tested classifiers using dense sentence embeddings generated from FastText. Sentence vectors were obtained by averaging token embeddings.

- **Custom FastText + Logistic Regression:** Trained from scratch on the project dataset (100-dimensions, 5 epochs).
- **Pretrained FastText + Logistic Regression:** Used vectors from `cc.en.300.vec`, a 300-dimensional pre-trained model trained on Common Crawl. Vectors were loaded with Gensim and stored using key-value format.

These approaches provided improved robustness compared to TF-IDF but still underperformed on free-form patient descriptions, with Top-1 real-world accuracy not exceeding 40%.

C. Transformer-Based Models

To address limitations in semantic representation, we fine-tuned several transformer models using Hugging Face’s `BertForSequenceClassification` architecture [20]. Each model was trained with stratified 80-10-10 train-validation-test splits and evaluated on validation, internal test, and real-world datasets.

The following transformer models were tested:

- **BERT-base** (uncased) [8]
- **ClinicalBERT** (pretrained on MIMIC-III clinical notes) [10]
- **BioBERT** (pretrained on biomedical literature) [9]
- **PubMedBERT** (trained from scratch on PubMed abstracts and full text) [11]
- **BioMedBERT** (fine-tuned biomedical variant with enhanced clinical performance) [16]
- **RoBERTa-base** (robustly optimized BERT, trained on more data with dynamic masking) [12]

Each model was trained using three input variations:

- 1) **Text Only:** Cleaned symptom description
- 2) **NER Only:** NER-extracted entities only
- 3) **Text + NER:** Combined input of symptom text and extracted entities

This variety of architectures and input formats allowed us to comprehensively evaluate the impact of linguistic richness and biomedical structure on disease classification performance. The final model selection and comparative results are detailed in the next section.

D. Training Configuration

All transformer models were trained using Hugging Face’s `Trainer` API [20] with the following settings:

- Learning rate: `2e-5`
- Epochs: 10
- Weight decay: `0.05`
- Batch size: 8
- Warmup ratio: `0.1`
- Scheduler: `cosine`
- Evaluation and model saving strategy: `epoch`
- Metric for best model: `accuracy`
- Load best model at end: `True`
- Logging steps: 50

These settings were chosen based on standard practices for fine-tuning transformer models in healthcare NLP, balancing performance and training stability.

VI. RESULTS

A. Evaluation Strategy

Each model was evaluated using three distinct datasets:

- **Validation Set (10%):** Used to select the best model checkpoint during training.
- **Test Set (10%):** Used for final in-domain evaluation after training.

TABLE I
EXAMPLES OF EXTRACTED ENTITIES FROM SYMPTOM DESCRIPTIONS

Symptom Description	Extracted Entities
I have been experiencing a skin rash on my arms, legs, and torso for the past few weeks. It is red, itchy, and covered in dry, scaly patches.	skin, rash, torso, itchy, dry, scaly
Flare-ups get worse in winter, and my scalp gets thick, crusty plaques.	flareups, worse, scalp, crusty
I have painful plaques on my knees and back that get worse in winter.	painful, plaques, knees, back, worse
My nails have small dents or pits in them, and they often feel inflammatory and tender to the touch. Even there are minor rashes on my arms.	nails, dents, pits, inflammatory, tender, rash
I lost my voice and my throat feels scratchy with a light cough.	lost, voice, throat, scratchy
There are clusters of pimples on my cheeks that leave dark marks afterward.	clusters, pimples, cheeks, dark marks

- **Simulated patient-style Evaluation:** All models were initially tested on a small benchmark of six free-form symptom descriptions. These examples were carefully reviewed by physicians to ensure clinical accuracy and represent realistic patient presentations. Based on this initial performance, the three top-performing transformer models—RoBERTa-base, BioMedBERT, and ClinicalBERT—were selected for more extensive Simulated patient-style testing.

A comprehensive 96-case test set was then constructed, containing four distinct symptom descriptions for each of the 24 disease labels. These descriptions were generated by multiple AI systems (Claude, GPT, Gemini, DeepSeek) and were clinically validated to ensure both realism and diversity in language, severity, and presentation patterns. This dataset was used exclusively to assess each model’s ability to generalize beyond the training distribution.

Performance was reported using:

- **Validation Accuracy (%)**: Accuracy on the internal validation set.
- **Test Accuracy (%)**: Accuracy on the held-out test set.
- **Simulated patient-style Top-1 Accuracy**: How often the model’s first prediction is the correct disease.
- **Simulated patient-style Top-3 Accuracy**: How often the correct disease appears among the model’s top three suggestions.

All transformer models were trained using a consistent configuration (detailed in Section V-D) to ensure fair comparison across architectures and evaluation conditions.

B. Performance on 6-Sample Benchmark

Table II summarizes the performance of all tested models on the 6-case Simulated patient-style benchmark set. To enhance interpretability across all experiments, Figure 3 presents a heatmap comparing validation, test, and Simulated patient-style Top-1/Top-3 accuracy across the entire set of classical and transformer models. The impact of both LLM-based augmentation and NER integration is clearly visible, with RoBERTa+NER achieving top performance across all metrics.

TABLE II
PERFORMANCE ON 6-SAMPLE SIMULATED PATIENT-STYLE BENCHMARK.

Model	Input Type	Val (%)	Test (%)	Top-1, Top-3
Logistic Regression	TF-IDF	96.0	98.3	2/6, 3/6
Random Forest	TF-IDF	95.0	96.6	2/6, 4/6
Logistic Regression	FastText (Custom)	80.0	77.0	1/6, 3/6
Random Forest	FastText (Pretrained)	85.0	78.0	3/6, 4/6
BERT-base	Text	95.0	96.0	2/6, 5/6
ClinicalBERT	Text	90.3	88.0	6/6, 6/6
ClinicalBERT	Text + NER	91.7	90.3	6/6, 6/6
BioBERT	Text	93.1	90.2	4/6, 6/6
BioBERT	Text + NER	93.8	91.0	6/6, 6/6
BiomedBERT	Text	94.4	89.6	5/6, 6/6
BiomedBERT	Text + NER	94.4	90.4	6/6, 6/6
PubMedBERT	Text	93.1	90.2	6/6, 6/6
PubMedBERT	Text + NER	94.4	85.4	6/6, 6/6
RoBERTa-base	Text	94.4	93.8	5/6, 6/6
RoBERTa-base	Text + NER	96.1	95.0	6/6, 6/6

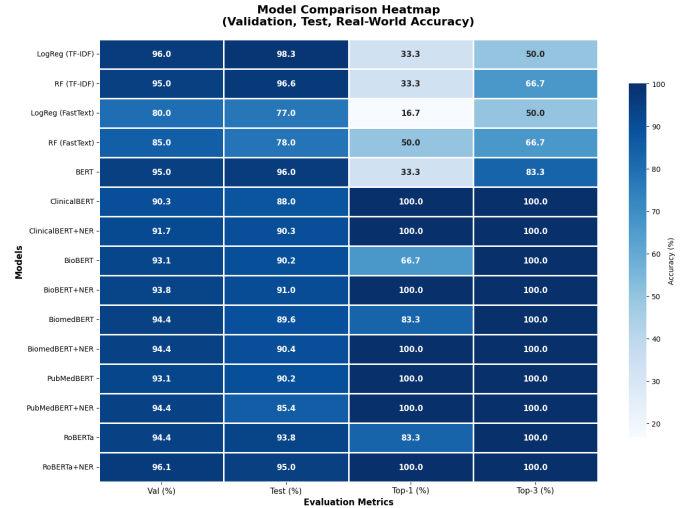


Fig. 3. Heatmap comparing validation, test, and Simulated patient-style Top-1/Top-3 accuracy across all tested models. NER integration and LLM-based augmentation significantly improved generalization, especially for transformer models. RoBERTa+NER achieved the highest overall performance.

C. Extended Simulated Patient-style Evaluation (96 Cases)

The three best-performing transformer models were further evaluated on a larger set of 96 Simulated patient-style symptom descriptions (four per disease). Table III presents their Top-1 and Top-3 accuracy scores.

RoBERTa-base showed the highest overall performance across all Simulated patient-style examples, confirming its robustness and generalization capability.

TABLE III
SIMULATED PATIENT-STYLE EVALUATION (96 DIVERSE SYMPTOM DESCRIPTIONS)

Model	Top-1 Accuracy (%)	Top-3 Accuracy (%)
ClinicalBERT (Text + NER)	87.50	100.00
RoBERTa-base (Text + NER)	95.83	100.00
BioMedBERT (Text + NER)	89.58	95.83

D. Final Model Selection

RoBERTa-base with Text + NER input achieved the strongest results across validation, test, and Simulated patient-style evaluations:

- **Validation Accuracy:** 96.13%
- **Test Accuracy:** 95.00%
- **Simulated patient-style Top-1 Accuracy:** 92/96 (95.83%)
- **Simulated patient-style Top-3 Accuracy:** 96/96 (100%)

However, these results are based on synthetic data and require validation with real patient populations before clinical application.

VII. DISCUSSION

The development of an accurate and generalizable disease classification model from free-text symptom descriptions required iterative experimentation, diagnostic failure analysis, and the strategic integration of advanced NLP techniques. Each phase of the project, whether yielding strong results or revealing limitations, offered crucial insights into the challenges of modeling clinical language variation.

Our initial approach, based on **TF-IDF vectorization** coupled with classical classifiers (Logistic Regression and Random Forest), achieved high internal performance (up to 98% accuracy). However, these models performed poorly on Simulated patient-style symptom narratives, correctly classifying only 2 out of 6 external test cases. This exposed a fundamental limitation of TF-IDF: its inability to capture semantic context. Since TF-IDF treats tokens as independent features without accounting for syntax, paraphrasing, or latent meaning, it fails to adapt when faced with phrasing variations common in real patient speech.

To address these limitations, we adopted **FastText embeddings**, which offer subword-level modeling and greater lexical flexibility. We evaluated both a custom-trained FastText model and a widely-used pretrained variant (`cc.en.300.vec`). However, despite theoretical advantages, the empirical gains were limited. Validation and test scores declined (to 77–80%), and Simulated patient-style performance remained static. These results indicate that while FastText improves over TF-IDF in terms of morphology and token coverage, sentence-level meaning is still poorly represented when using averaged word vectors.

This led us to explore **transformer-based models**, which provided a significant leap in performance. Unlike earlier techniques, transformers employ self-attention mechanisms to capture long-range dependencies and contextual relationships across tokens. Models such as BERT, ClinicalBERT,

BioBERT, PubMedBERT, and RoBERTa demonstrated substantial improvements, especially when evaluated on out-of-distribution patient-style inputs.

Notably, RoBERTa-base, a general-purpose language model, consistently outperformed domain-specific biomedical transformers across validation, internal test, and Simulated patient-style benchmarks. This was an unexpected outcome, as RoBERTa lacks any pretraining on medical literature. One plausible explanation is that RoBERTa’s exposure to a significantly broader and more diverse corpus, encompassing informal, colloquial, and dialogue-style language, enhanced its ability to understand symptom descriptions phrased in natural, patient-oriented terms. By contrast, biomedical models are optimized for structured clinical documentation (e.g., physician notes, PubMed abstracts), which may not resemble real-world patient communication.

However, even the strongest transformer models were initially constrained by the limited linguistic diversity of the training dataset. Each disease label was represented by only one phrasing pattern, which restricted the models’ ability to generalize. This highlighted a clear **data-centric bottleneck**, rather than a model-centric one.

To overcome this, we implemented **large language model (LLM)-based data augmentation**. Using systems such as ChatGPT, Claude, Gemini, and DeepSeek, we generated 10 additional patient-style descriptions per disease. These samples were manually reviewed for clinical accuracy. The results were transformative: the final model, RoBERTa with Text + NER, achieved a Top-1 accuracy of 95.83% on a 96-sample Simulated patient-style benchmark, compared to just 33–66% in early transformer experiments.

In parallel, we introduced **Named Entity Recognition (NER)** using a transformer-based biomedical NER model. The extracted entities were cleaned, normalized, and used either as standalone input or concatenated with the original symptom text. The hybrid approach (Text + NER) consistently outperformed both standalone representations, suggesting that fusing structured medical concepts with natural symptom phrasing creates a richer, more discriminative input representation. In summary, our experiments revealed several key principles:

- Surface-level vectorization methods (TF-IDF) are insufficient for capturing clinical meaning in real-world scenarios.
- Pretrained word embeddings (FastText) offer only incremental gains unless contextualized at the sentence level.
- Transformer architectures offer powerful semantic modeling capabilities, but only perform optimally when supported by representative, diverse data.
- RoBERTa, despite not being domain-specific, can outperform biomedical transformers when generalization to informal language is needed.
- The integration of structured NER information with natural language input enhances robustness and interpretability.

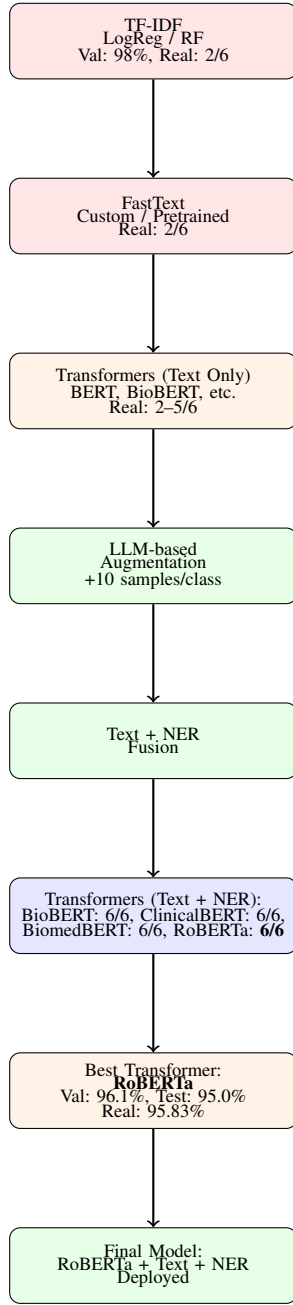


Fig. 4. Compact pipeline showing the transition from traditional methods to final RoBERTa-based model. Data augmentation and NER were key to unlocking generalization.

Ultimately, the combination of a semantically rich model (RoBERTa), diverse Simulated patient-style symptom data, and structured NER-based augmentation allowed us to close the gap between lab accuracy and real-world usability. Each failed approach revealed a limitation, while each successful adaptation pointed toward a scalable and clinically relevant solution. As shown in Figure 4, this evolutionary path reveals how data-centric and model-centric innovations jointly enabled robust generalization, a blueprint for future diagnostic NLP systems.

VIII. LIMITATIONS

Despite the strong performance of our RoBERTa-based classification pipeline, several limitations must be acknowledged, each of which may affect the generalizability, scalability, or clinical deployment of the system.

- **Limited Dataset Size and Linguistic Diversity:** Although we expanded the dataset to 1,440 samples through LLM-based augmentation, this size remains modest relative to standard requirements for training large-scale transformers. Moreover, the dataset may not fully capture the linguistic diversity found across different patient populations, dialects, or health literacy levels.
- **Synthetic Augmentation Bias:** The additional symptom descriptions were generated by language models and manually verified for medical accuracy. However, such synthetic data may still reflect stylistic or semantic biases inherent in the generative systems. This could limit generalization to symptom expressions that differ significantly in style or cultural context.
- **Monolingual Focus:** The current system supports only English inputs. In practice, symptom articulation varies significantly across languages and cultures. The lack of multilingual support limits the global applicability of the model, especially in diverse or multilingual healthcare environments.
- **NER Sensitivity to Informal Language:** While transformer-based NER improved robustness, entity extraction remains sensitive to informal phrasing, abbreviations, and misspellings commonly used by patients. Some medically relevant phrases may be missed or fragmented, weakening the structured signal.
- **Lack of Clinical Validation:** This system has not undergone clinical validation with real patients or healthcare providers. Therefore, issues such as usability, safety, and clinician interpretability remain open challenges. Real-world deployment would require ethical approvals, user studies, and ongoing monitoring.
- **Limited Model Transparency:** Like most deep learning systems, the RoBERTa-based model operates as a black box. Without additional interpretability tools (e.g., attention heatmaps, saliency maps, or rationale extraction), the model's decisions may be difficult to explain to clinicians or patients, potentially limiting trust in critical scenarios.

IX. CONCLUSION AND FUTURE WORK

This study demonstrates the viability of building a robust, generalizable, and deployable disease classification model from free-text symptom descriptions using a hybrid NLP pipeline. We systematically explored traditional vectorization techniques (TF-IDF), subword embedding methods (FastText), and transformer-based models trained on both unstructured and structured inputs.

Initial models showed high internal accuracy but failed to generalize to patient-style symptom descriptions. This gap highlighted the importance of both semantic modeling and data

diversity. The integration of LLM-generated symptom augmentations and transformer-based biomedical Named Entity Recognition significantly improved performance, enabling the model to handle real-world linguistic variability.

Our final model, RoBERTa with combined Text + (NER) input, achieved 96.1% validation accuracy, 95.0% test accuracy, and 95.83% Top-1 accuracy across 96 clinically validated, real-world symptom cases. These results confirm that semantic richness, biomedical structure, and language variation are all critical for developing robust NLP systems for healthcare applications.

Importantly, we successfully deployed the model within a real-time web interface, enabling users to input natural symptom descriptions and receive a likely diagnosis, condition summary, and follow-up guidance. This integration represents a meaningful step toward real-world applicability in telemedicine and patient-facing digital tools.

Looking forward, several avenues remain open for future work:

- **Dataset Expansion:** We plan to incorporate real patient narratives from health forums, clinical records, or crowd-sourced data to further improve diversity and realism.
- **Multilingual Support:** Extending the model to support multiple languages will enhance accessibility in global healthcare contexts.
- **Incorporating Clinical Metadata:** Adding patient-specific variables (e.g., age, gender, vitals) could further improve classification accuracy and support personalized diagnostics.
- **Improving Interpretability:** Developing tools for visualizing model attention, highlighting relevant symptoms, or generating justifications could increase clinical trust and transparency.
- **Clinical Evaluation:** The next step involves deploying the system in real-world environments, e.g., triage systems, clinics, or mobile health apps, to evaluate safety, effectiveness, and user satisfaction under operational conditions.

This work represents a proof-of-concept that requires extensive clinical validation before practical use. In summary, this research offers a scalable framework for building AI-powered diagnostic tools from unstructured symptom narratives. By combining semantic depth, biomedical structure, and language diversity, our system moves closer to bridging the gap between machine learning performance and potential healthcare utility.

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